

COMP4616 – Fundamentals of Machine Learning

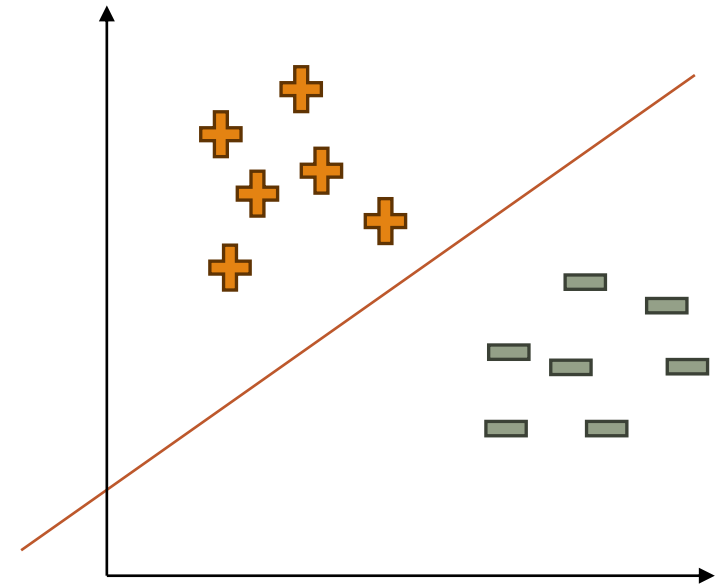
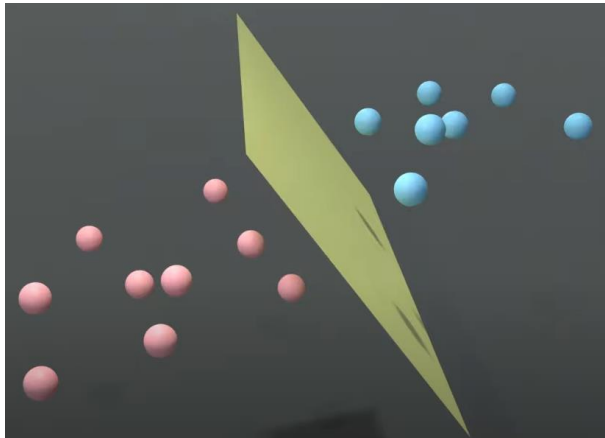
ASST. PROF. TUĞBA ERKOÇ

SPRING 2026

WEEK 6: SUPPORT VECTOR MACHINES (SVM)

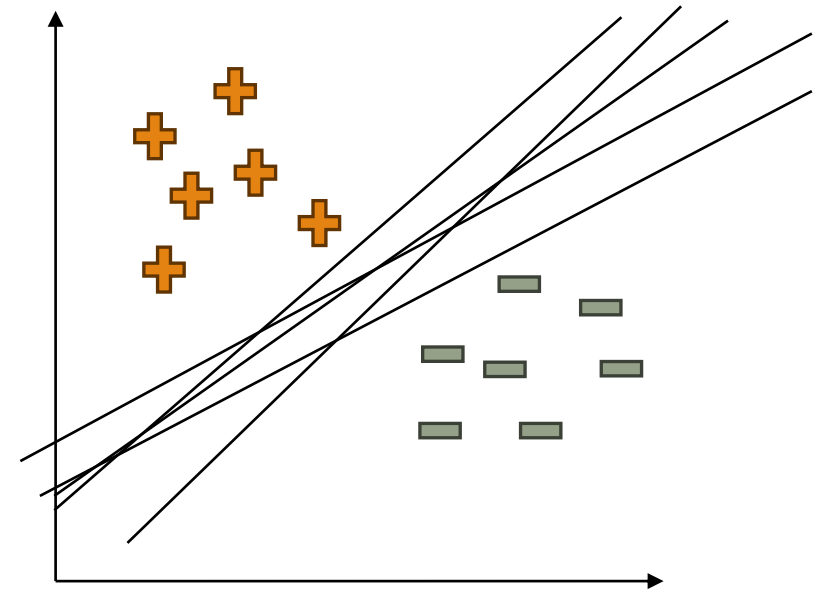
Support Vector Machines(SVM)

- **Support Vector Machines(SVM)** is a **supervised** learning algorithm.
- It is another **linear classification** method
- SVM tries to find the **best hyperplane** that separates positive and negative examples in data in the feature space (linearly separable patterns)
 - In **2D** space, it is a **line**
 - In **3D** and higher space, it can be a **plane**



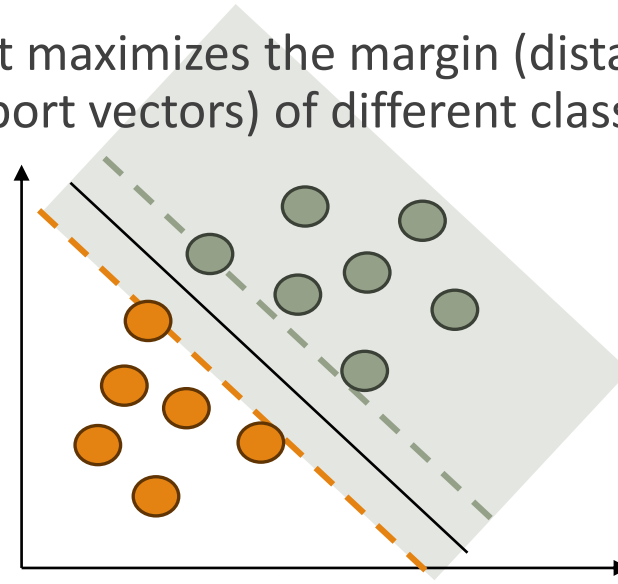
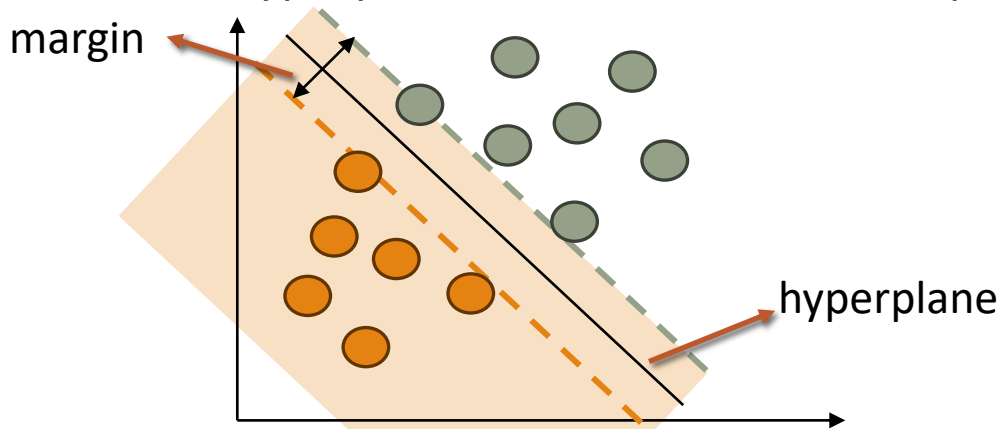
Support Vector Machines(SVM)

- General idea is to find a line that separates data into two classes:
 - Start by randomly picking with a line
 - Pick random data points and check whether the line is separating the points well or not.
 - If the line is not separating the points well, we move the line until we find the line which correctly separates the points.
 - Moving the line means changing the equation of the line
 - We repeat these until we are satisfied.
 - i.e. generalization error is minimized.



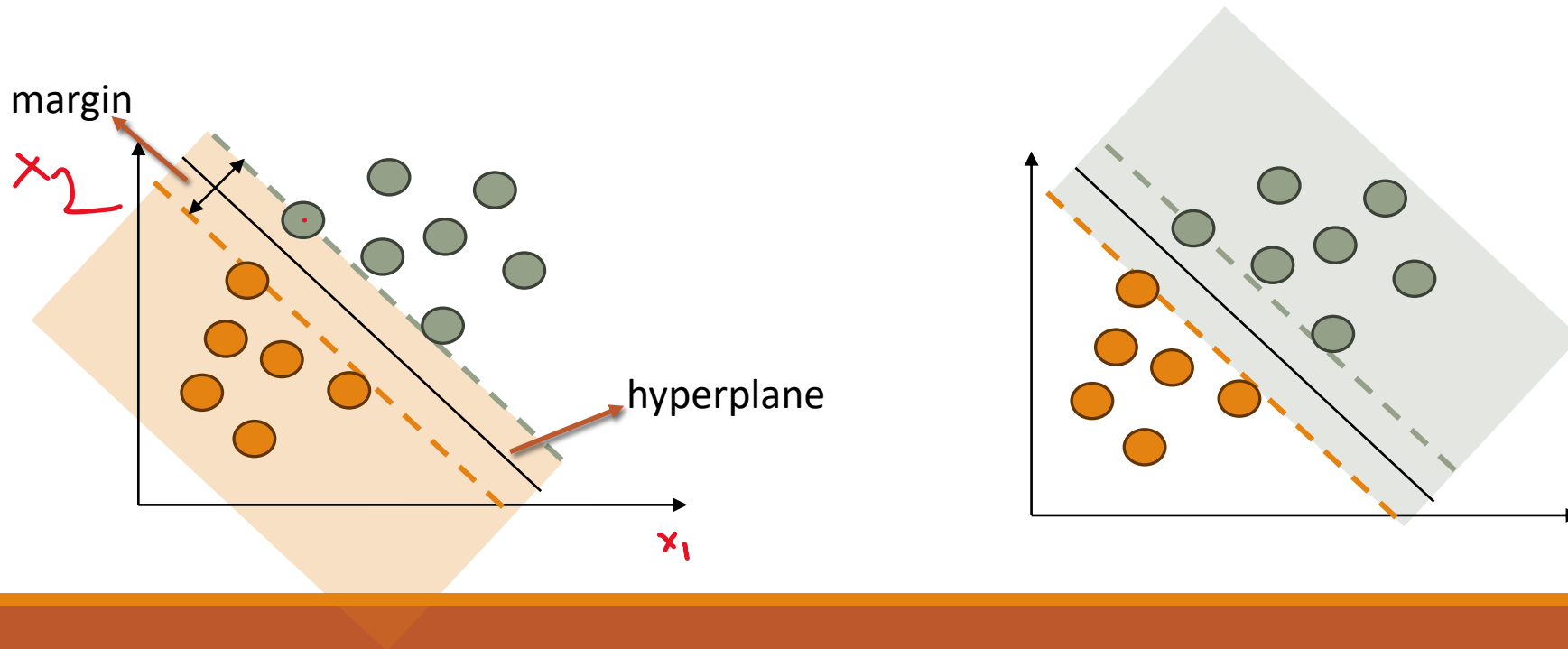
Support Vector Machines(SVM)

- SVM adopts a similar approach. However, the way to pick the best line depends on a concept called **margin**.
- Margin is the smallest distance between the **decision boundary** and the **closest samples** to the decision boundary.
- SVMs aim to find the optimal hyperplane that maximizes the margin (distance) between the hyperplane and the nearest data points (support vectors) of different classes.



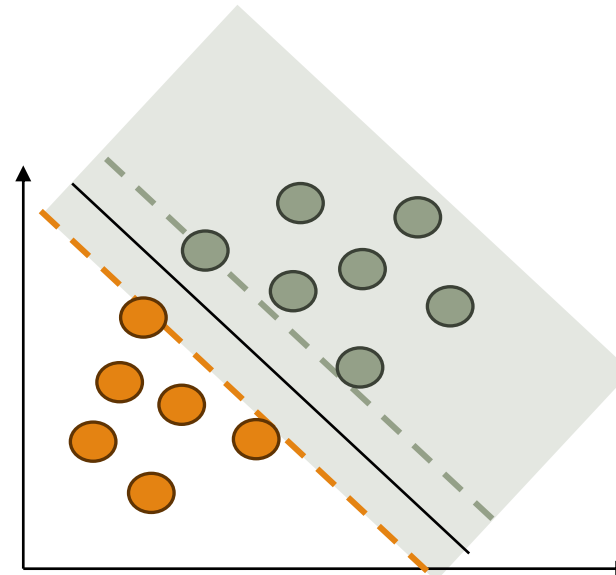
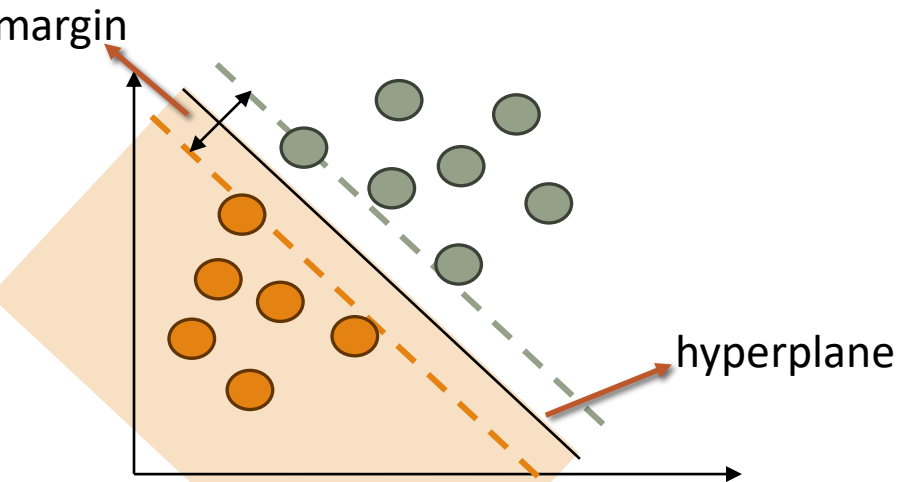
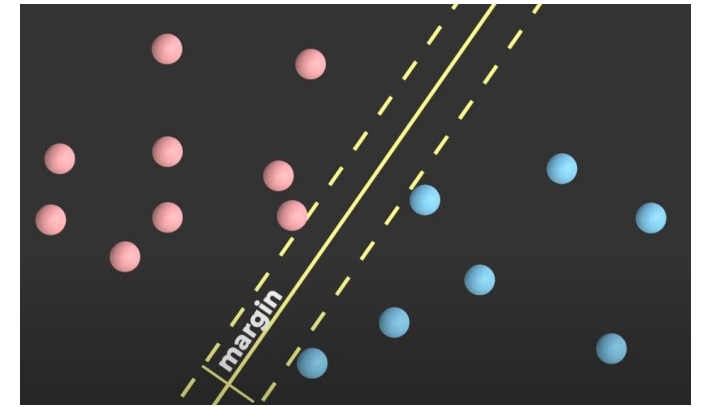
Support Vector Machines(SVM)

- SVM tries to find the hyperplane that maximizes the distance between two categories(classes).
 - The objective of the SVM is to find the optimal separating hyperplane which maximizes the margin of the training data.
- The optimal hyperplane is the one with **largest margin**.



Support Vectors

- The **data points** that falls **on the margin** are called **support vectors**.
- They are the data points **most difficult to classify**
- They influence the **position** and **orientation** of the hyperplane

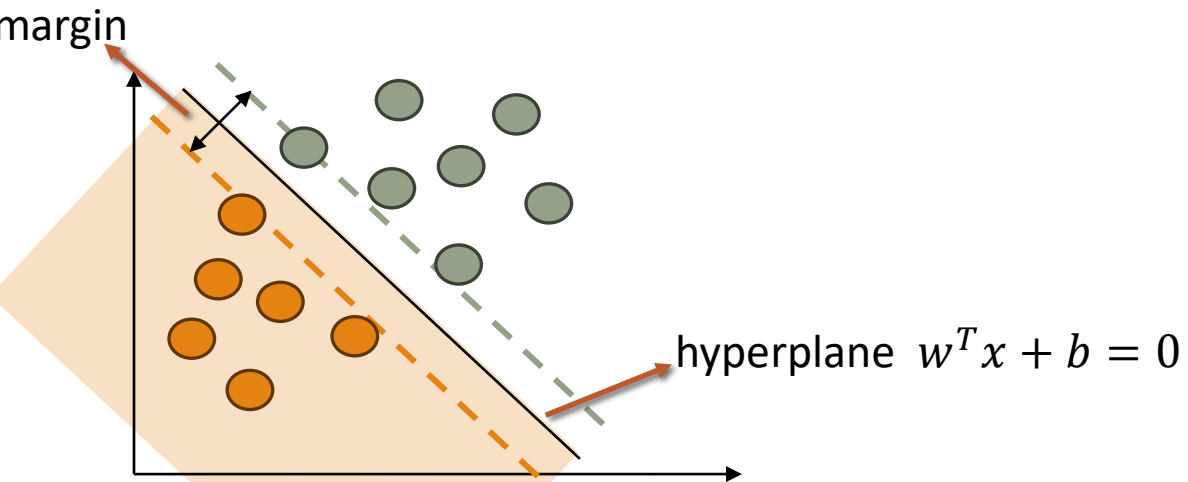


Decision Boundary

- The **decision boundary** is the hyperplane represented by the line:

$$w^T x + b = 0$$

- It is the center line that separates the two classes of data.
- It does not pass through any data points. It lies **equidistant** between the two closest points (support vectors) from each class



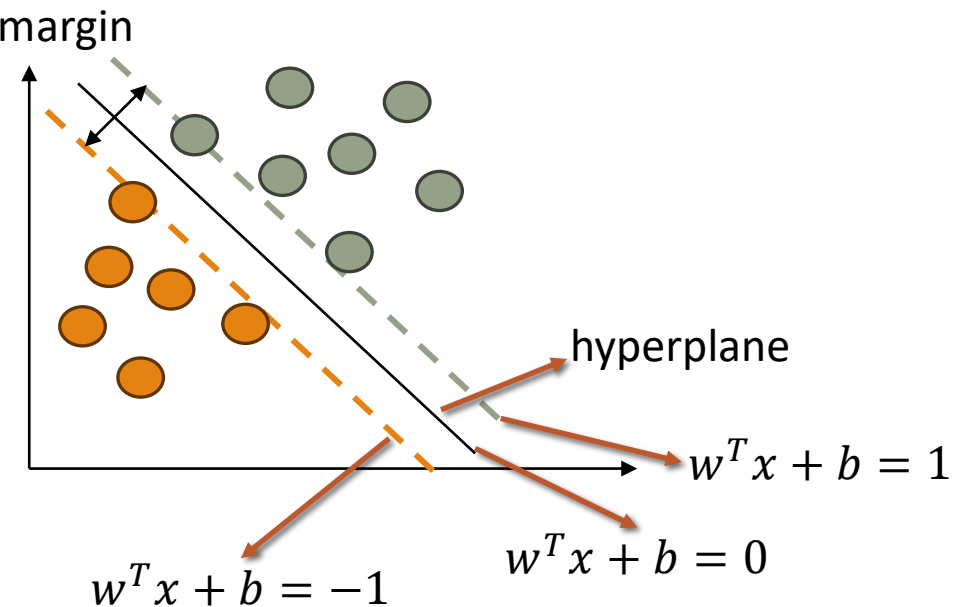
The Margin Lines

- The two lines parallel to the decision boundary are the **margin boundaries**:

$$w^T x + b = 1$$

$$w^T x + b = -1$$

- The distance between them is the margin, and the support vectors lie on these lines.



Support Vector Machines(SVM) - The Math Behind the Margin

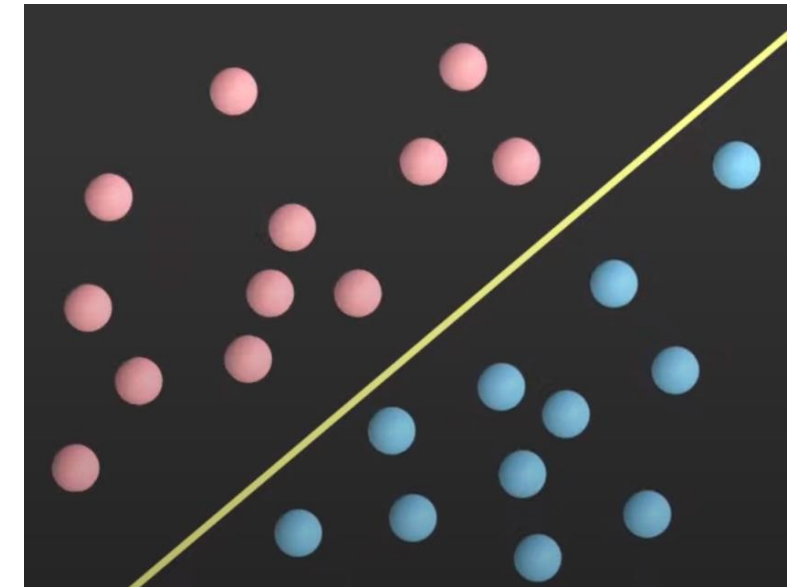
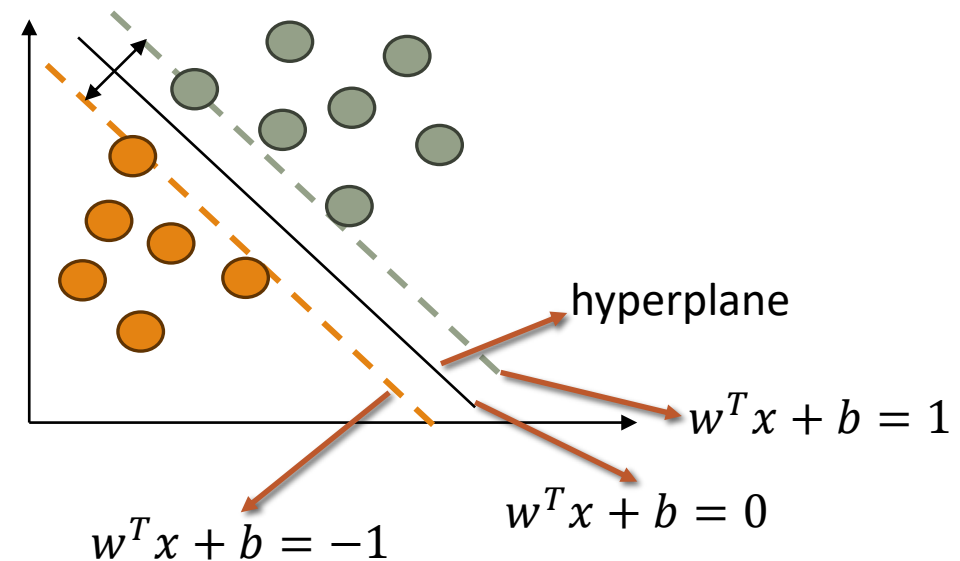
- Maximizing the margin is a **convex optimization problem** where each data point belonging to categories should fall on the correct side of the hyperplane.
- Equation of the hyperplane is defined as

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \leftarrow w^T x + b = 0$$

bias

we are guaranteed to find the best possible boundary

$$\begin{aligned} w^T x + b &\geq 1 \quad \forall x \in c_1 \\ w^T x + b &\leq -1 \quad \forall x \in c_2 \end{aligned}$$



Support Vector Machines(SVM)

- If we assume the equation of the hyperplane line is

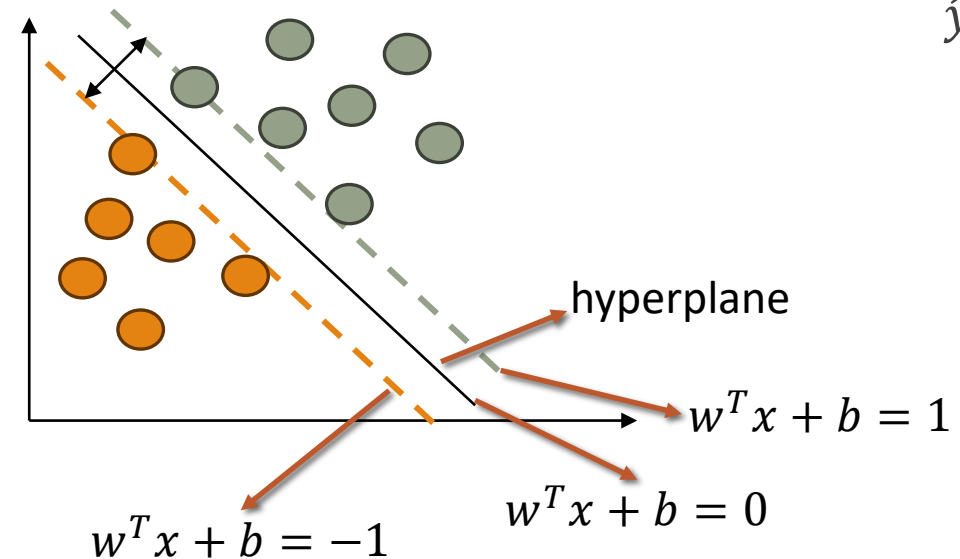
$$w^T x + b = 0$$

$$ax_1 + bx_2 + c = 0$$

- then

$$\hat{y} = \begin{cases} +1 & \text{if } ax_1 + bx_2 + c \geq 0 \\ -1 & \text{if } ax_1 + bx_2 + c < 0 \end{cases}$$

- If a point falls on or above the decision boundary: classify as +1 (positive class).
- If it falls below the line: classify as -1 (negative class).



The margin — the distance between the two margin lines — is given as:

$$E_{margin} = \frac{2}{\sqrt{a^2 + b^2}} \longrightarrow \text{Norm of the vector}[a,b]$$

Support Vector Machines(SVM)

- The margin — the distance between the two margin lines — is given as:

$$E_{margin} = \frac{2}{\sqrt{a^2+b^2}} = \frac{2}{\|w\|}$$

- The margin gets smaller if the weight vector has a larger magnitude, and larger if the weight vector is shorter.
- Thus, the **length of w influences the margin width**
- SVM's goal: maximize the margin, which mathematically means minimizing $\|w\|^2$ subject to correct classification constraints.
 - A bigger margin $\rightarrow$ more robust $\rightarrow$ better generalization

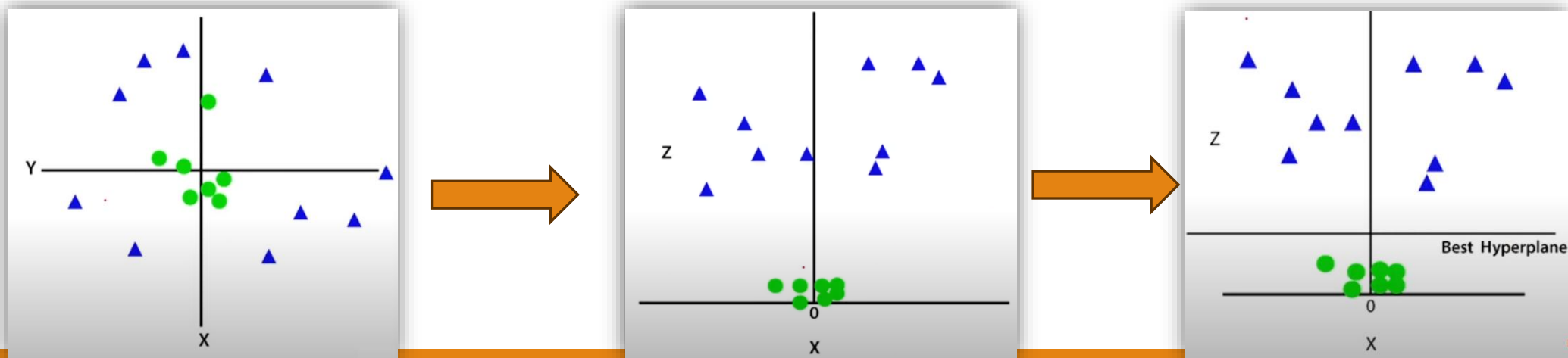


Optimization algorithm
involves derivatives.
This is differentiable and
smooth everywhere

Support Vector Machine

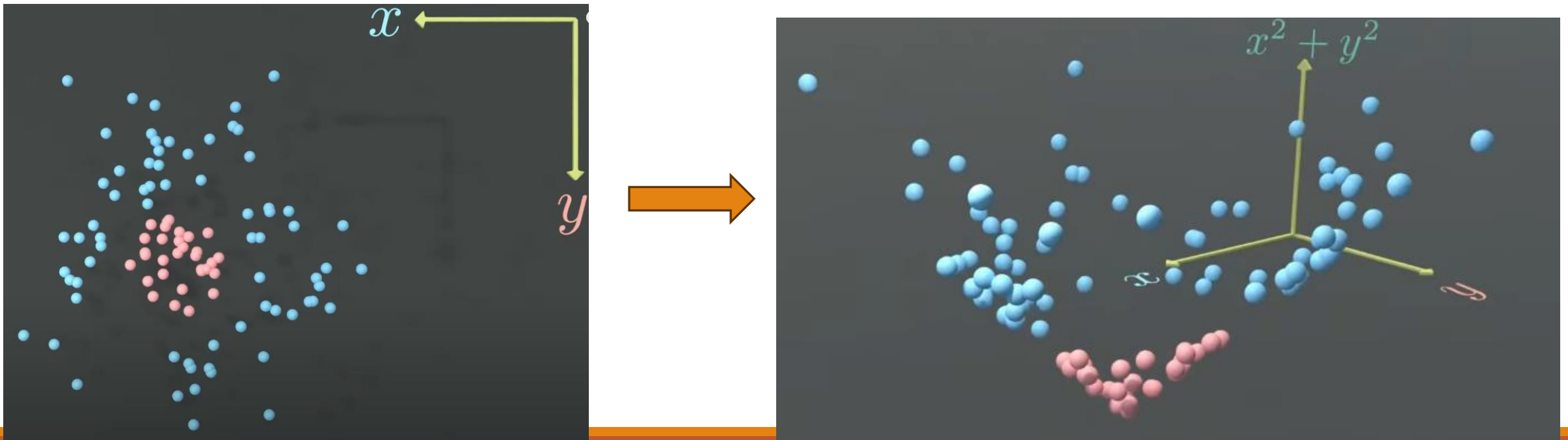
- SVM can work with small data.
- Application areas include
 - Face detection, spam mail detection, text recognition.
- SVM algorithm can be extended to patterns that are not linearly separable by transforming the original data into a new space by use of **kernel functions**.
- A mapping function like $z = x^2 + y^2$ can be used to add a third dimension to the data

kernel functions: they let you work in high-dimensional spaces without explicitly computing the transformation.



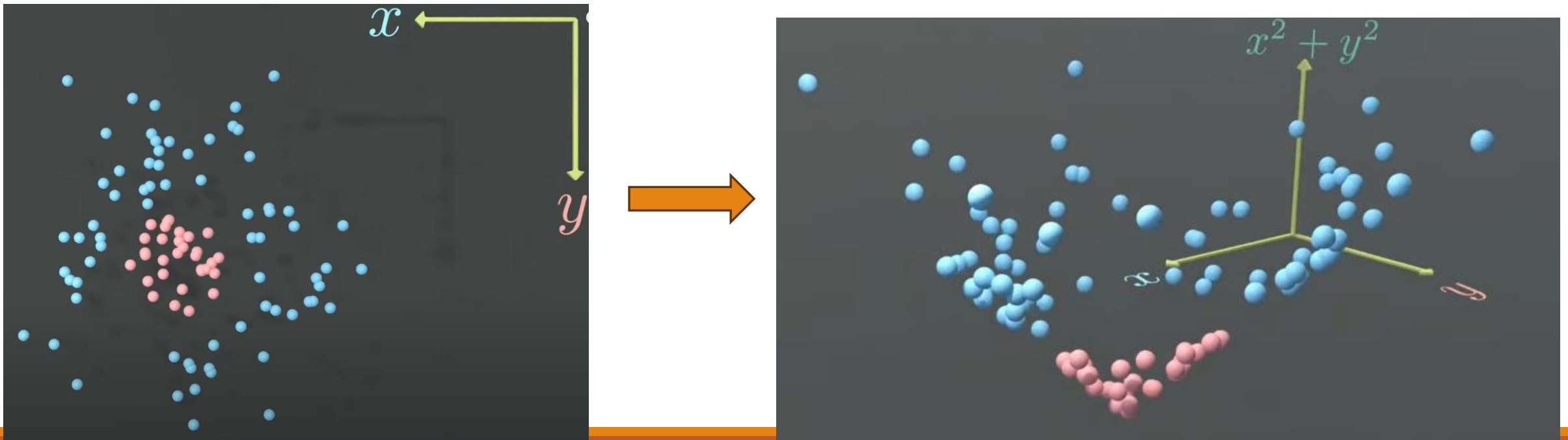
Support Vector Machine

- The two classes (e.g., blue and pink dots) are not linearly separable. You can't draw a straight line to cleanly separate them. This is a classic case where a linear SVM would struggle.
- A nonlinear mapping function like $z = x^2 + y^2$ (polynomial kernel) can be used to add a third dimension to the data



Support Vector Machine

- After the mapping to 3D feature space with the kernel trick, a linear plane (hyperplane in 3D) can easily separate these two classes.
 - In 2D, the data was entangled.
 - But in 3D, the data becomes linearly separable



Support Vector Machine Example

- For some sample data, we can apply **SVM** to find out which class the data point **(2,4)** belongs to by using **scikit-learn**.
 - Train on a small 2D dataset with binary labels (positive/negative)
 - Fit a linear SVM
 - Predict the class of a new, unseen data point: (2,4)

```
sample:[4.36722546 3.91135913] label: 1
sample:[3.56655455 3.35579053] label: 1
sample:[3.11554942 3.80246008] label: 1
sample:[2.85994882 3.26756178] label: 1
sample:[3.81881573 2.8796789 ] label: 1
sample:[3.05019265 2.82296865] label: 1
sample:[3.19714802 2.92841881] label: 1
sample:[2.33669911 3.16265878] label: 1
sample:[2.23387983 3.47421502] label: 1
sample:[3.93875757 2.77107661] label: 1
sample:[3.55342478 2.56034364] label: 1
sample:[3.50979167 2.96120061] label: 1
sample:[2.7988496 1.6727912] label: 1
sample:[3.43267969 3.62780505] label: 1
sample:[3.43281998 3.26717261] label: 1
sample:[2.74640945 3.62994616] label: 1
sample:[2.56545997 2.40119278] label: 1
sample:[3.37555692 2.71000295] label: 1
sample:[2.85640412 3.6348177 ] label: 1
sample:[3.39813856 2.2073319 ] label: 1
sample:[ 0.14026867 -0.7192688 ] label: -1
sample:[ 0.15493099 -0.39118269] label: -1
sample:[-0.85702197 -0.31802534] label: -1
sample:[0.16017117 0.25807394] label: -1
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sample:[-0.12876035 -0.14574701] label: -1
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sample:[-0.32234462 0.47452769] label: -1
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